

PATENT LANDSCAPE FOR HYDROGENATED NITRILE RUBBER (HNBR) SYNTHESIS, MODIFICATION, AND FORMULATION

1. ABSTRACT

This report surveys a focused patent set relevant to hydrogenated nitrile rubber (HNBR), defined as nitrile-butadiene rubber (NBR) in which the diene-derived unsaturation has been selectively reduced to improve heat, ozone, oil, and aging resistance. The selected patents span three major technical themes: direct hydrogenation of NBR to HNBR, post-hydrogenation purification and molecular-weight control, and downstream compounding/formulation strategies that exploit HNBR's saturated backbone. Across the landscape, the core synthetic route remains solution-phase hydrogenation of NBR using homogeneous transition-metal catalysts, especially rhodium- and ruthenium-based systems, with process emphasis on catalyst choice, hydrogen pressure, temperature, and residual double-bond control. Several patents also address simultaneous metathesis/hydrogenation or metathesis-driven molecular-weight reduction to produce low-Mooney HNBR with improved processability.

The formulation patents show that HNBR performance is strongly tuned by filler package, peroxide cure system, and co-crosslinking agents. One cluster targets high-temperature vulcanizates using microgels, unsaturated carboxylic acids/salts, and peroxides to improve modulus retention and demolding behavior. Another cluster focuses on intake-manifold and sealing applications, balancing acrylonitrile content, iodine number, ester plasticizer level, peroxide loading, and optional polyfunctional unsaturated compounds to optimize oil/fuel resistance and low-temperature flexibility. A separate composition strategy blends HNBR with unhydrogenated NBR to reduce cost while maintaining sealing performance, and another patent removes ruthenium catalyst residues from optionally hydrogenated nitrile rubber using functionalized ion-exchange resins.

Overall, the evidence indicates that the most robust HNBR synthesis routes are homogeneous hydrogenation of NBR in solution, often with selective catalysts and controlled hydrogenation to high conversion of diene units, followed by purification to reduce residual metal. For application-oriented HNBR compounds, the most consistent trends are peroxide curing, careful control of residual unsaturation/iodine value, and filler/plasticizer optimization to balance heat resistance, oil resistance, cold flexibility, and processability.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US9321910B2
- Patent Title: Microgel-containing vulcanizable composition based on hydrogenated nitrile ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent does not disclose HNBR synthesis itself; it discloses vulcanizable compositions based on hydrogenated nitrile rubber, unsaturated carboxylic acid or salt, peroxide, and microgels. The technical contribution is formulation design for improved high-temperature vulcanizate performance rather than polymer preparation. - Hydrogenated nitrile rubber degree of hydrogenation: 95 to 100% — Claim 1 - Hydrogenated nitrile rubber Mooney viscosity (ML 1+4 @ 100°C): 10 to 120 MU — Claim 1 - Unsaturated carboxylic acid/salt loading: 1 to 100 parts by weight per 100 parts rubber — Claim 1 - Peroxide loading: 0.2 to 8 parts by weight per 100 parts rubber — Claim 1 - Microgel glass transition temperature: less than or equal to -20°C — Claim 1 - Microgel loading: up to 60 parts by weight per 100 parts rubber — Claim 2 - Microgel glass transition temperature: less than or equal to -50°C — Claim 3 - Hydrogenated nitrile rubber Mooney viscosity (ML 1+4 @ 100°C): 15 to 100 MU — Claim 4 - Operating temperature: 130°C — Background/Summary - Hot air storage temperature: 150°C — Summary

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance Relevant as a downstream HNBR compounding patent that shows how hydrogenated nitrile rubber can be formulated for high-temperature durability. It is useful for understanding cure packages and filler/microgel strategies that preserve HNBR properties after hydrogenation.

Patent 2

- Patent Number: US7919563B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed

- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent prepares low-molecular-weight HNBR by subjecting nitrile rubber to a metathesis reaction in the presence of ruthenium/osmium metathesis catalysts, optionally with a co-olefin and solvent. The process is designed to reduce molecular weight and narrow molecular-weight distribution while retaining the HNBR backbone after hydrogenation-related processing. - Target molecular weight (MW): 30,000 to 250,000 — Claim 1 - Target Mooney viscosity (ML 1+4 @ 100°C): 3 to 50 — Claim 1 - Target polydispersity index: less than 2.5 — Claim 1 - Cement concentration: 15 wt.-% solid — Example 1 - Ethylene pressure: 6.9×10^5 Pa — Example 1 - Reactor temperature: 80°C — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Co-olefin: 1-hexene — Example 4 - Co-olefin amount: 150 g — Example 4 - Mooney viscosity (ML 1+4 @ 100°C): 73 — Starting polymer (Table 2) - Mooney viscosity (ML 1+4 @ 100°C): 43 — Experiment 1 (Table 2) - Mooney viscosity (ML 1+4 @ 100°C): 28 — Experiment 2 (Table 2) - Mooney viscosity (ML 1+4 @ 100°C): 3 — Experiment 3 (Table 2) - Mooney viscosity (ML 1+4 @ 100°C): 31 — Experiment 4 (Table 2) - Mn: 85 kg/mol — Substrate (Table 2) - Mw: 296 kg/mol — Substrate (Table 2) - PDI: 3.50 — Substrate (Table 2) - Mn: 27 kg/mol — Typical product description - Mw: 54 kg/mol — Typical product description - PDI: 2.0 — Typical product description

Relevant Experimental Evidence - Example 1: Ethylene-treated cement at 80°C with 0.05 phr catalyst gave reduced molecular weight and Mooney viscosity. - Example 2: Increasing catalyst load to 0.10 phr further reduced Mooney viscosity relative to Example 1. - Example 3: 0.25 phr catalyst produced the lowest Mooney viscosity and narrowest distribution among the ethylene runs. - Example 4: 1-hexene co-olefin was used instead of ethylene, showing the process also works with a different chain-transfer olefin.

Technical Relevance Highly relevant because it provides a route to low-Mooney HNBR, which is important when conventional hydrogenation yields polymers that are too viscous for processing. It also demonstrates how metathesis/cross-metathesis can be used to tune HNBR molecular weight before or during hydrogenation-related processing.

Patent 3

- Patent Number: US7772305B2
- Patent Title: Hydrogenated NBR composition
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses HNBR-based sealing compositions rather than HNBR synthesis. The HNBR is compounded with ester plasticizer, organic peroxide, and optionally polyfunctional unsaturated compound to produce molded articles with improved oil, fuel, heat, and cold resistance. - Hydrogenated NBR acrylonitrile content: 25 to 44 wt.% — Claim 1 - Hydrogenated NBR iodine number: 40 to 65 — Claim 1 - Ester-based plasticizer loading: 2 to 23 parts by weight — Claim 1 - Organic peroxide loading: 0.5 to 10 parts by weight — Claim 1 - Polyfunctional unsaturated compound loading: 0.5 to 10 parts by weight — Claim 3 - Carbon black/white carbon loading: 20 to 150 parts by weight — Claim 4 - Preferred hydrogenated NBR acrylonitrile content: 30 to 40 wt.% — Claim 2 - Preferred ester-based plasticizer loading: 7 to 17 parts by weight — Claim 2 - Preferred organic peroxide loading: 4 to 8 parts by weight — Claim 2 - Press vulcanization temperature: 170°C — Example 1 - Press vulcanization time: 20 minutes — Example 1 - Oven vulcanization temperature: 160°C — Example 1 - Oven vulcanization time: 3 hours — Example 1 - Hydrogenated NBR AN content: 36 wt.% — Example 1 - Hydrogenated NBR iodine number: 28 — Example 1 and Comparative Example 1 - NBR AN content: 33 wt.% — Example 1 - White carbon loading: 34 parts by weight — Example 1 - Silane coupling agent loading: 2 parts by weight — Example 1 - Ester-based plasticizer loading: 15 parts by weight — Example 1 - Organic peroxide loading: 6 parts by weight — Example 1 - Hydrogenated NBR AN content: 36 wt.% — Example 2/3/4 blends - Blend iodine number: 34 — Example 2 - Blend iodine number: 40 — Example 3 - Blend iodine number: 48 — Example 4 - Hydrogenated NBR AN content: 36 wt.% — Comparative Example 2/3 - Hydrogenated NBR AN content: 44 wt.% — Comparative Example 4 - Hydrogenated NBR AN content: 25 wt.% — Comparative Example 5

Relevant Experimental Evidence - Example 1: A hydrogenated NBR/NBR blend with white carbon and silane coupling agent was compounded and peroxide-cured for sealing applications. - Example 2: Lower HNBR fraction increased blend iodine number to 34 while maintaining the same cure package. - Example 3: Intermediate blend ratio gave iodine number 40 and similar molded-product evaluation. - Example 4: Higher NBR fraction gave iodine number 48, showing the cost-reduction strategy via blending. - Comparative Example 1: Pure HNBR formulation served as the baseline for property comparison.

Technical Relevance Relevant because it defines practical HNBR composition windows for oil- and fuel-resistant molded parts. Although not a synthesis patent, it gives explicit HNBR property targets and cure conditions that are directly tied to the performance of hydrogenated NBR materials.

Patent 4

- Patent Number: US7846995B2
- Patent Title: ... -containing catalyst residues from optionally hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown

- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is not a synthesis patent for HNBR; it discloses removal of ruthenium-containing catalyst residues from optionally hydrogenated nitrile rubber solutions using functionalized ion-exchange resins. The process is a purification step applied after metathesis/hydrogenation chemistry to obtain low-ruthenium HNBR. - Ion-exchange resin particle diameter: 0.2 to 2.5 mm dry basis — Claim 1 - Ruthenium content in solution: 5 to 1000 ppm — Claim 2 - Ruthenium content in solution: 5 to 500 ppm — Claim 3 - HNBR solution concentration: 0.5 to 30 wt.% — Claim 4 - Treatment temperature: ca. 100°C — Example 2A - Treatment time: 66 hours — Example 2A - Rubber solution: 180 g standard nitrile rubber solution — Example 2A - Ion-exchange resin: 0.5 g Lewatit® MonoPlus MP 500 — Example 2A - Ru content after treatment: 10 ppm — Example 2A - Ru removal: 33% — Example 2A

Relevant Experimental Evidence - Example 2A: Functionalized resin treatment reduced Ru content in the rubber to 10 ppm. - Example 2B: Untreated control sample established the initial Ru baseline for comparison. - Example 1: The example table indicates the resin and process concept for catalyst residue removal.

Technical Relevance Highly relevant as a downstream purification step for HNBR made by ruthenium-catalyzed metathesis/hydrogenation routes. Removing residual Ru is important for color, stability, and commercial acceptability of hydrogenated nitrile rubber.

Patent 5

- Patent Number: US20090258980A1
- Patent Title: Hydrogenated nbr composition
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2009
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses HNBR/NBR blend compositions rather than HNBR synthesis. The formulation uses hydrogenated NBR blended with unhydrogenated NBR, ester plasticizer, peroxide, and optional polyfunctional unsaturated compound to balance cost, oil resistance, fuel resistance, and low-temperature properties. - Hydrogenated NBR acrylonitrile content: 25 to 44 wt.% — Claim 1 - Hydrogenated NBR iodine number: 32 to 65 — Claim 1 - Blend iodine number: 40 to 160 — Claim 1 - Ester-based plasticizer loading: 2 to 23 parts by weight — Claim 1 - Organic peroxide loading: 0.5 to 10 parts by weight — Claim 1 - Polyfunctional unsaturated compound loading: 0.5 to 10 parts by weight — Claim 1 - Blend composition: 95 to 55 wt.% hydrogenated NBR and 5 to 45 wt.% NBR — Claim 2 - Blend

composition: 85 to 65 wt.% hydrogenated NBR and 15 to 35 wt.% NBR — Claim 3 - Blend composition: 85 to 75 wt.% hydrogenated NBR and 15 to 25 wt.% NBR — Claim 4 - Preferred blend iodine number: 50 to 140 — Claim 5 - Preferred blend iodine number: 70 to 110 — Claim 6 - Carbon black/white carbon loading: 20 to 150 parts by weight — Claim 7 - Hydrogenated NBR AN content: 36 wt.% — Example 1 - Hydrogenated NBR iodine number: 28 — Example 1 - NBR AN content: 33 wt.% — Example 1 - White carbon loading: 34 parts by weight — Example 1 - Silane coupling agent loading: 2 parts by weight — Example 1 - Ester-based plasticizer loading: 15 parts by weight — Example 1 - Organic peroxide loading: 6 parts by weight — Example 1 - Hydrogenated NBR fraction: 90 to 60 parts by weight — Example 1 to 4 - NBR fraction: 10 to 40 parts by weight — Example 1 to 4 - Blend iodine number: 81 — Example 1 - Blend iodine number: 106 — Example 2 - Blend iodine number: 130 — Example 3 - Blend iodine number: 154 — Example 4 - Press vulcanization temperature: 170°C — Example 1 - Press vulcanization time: 20 minutes — Example 1 - Oven vulcanization temperature: 160°C — Example 1 - Oven vulcanization time: 3 hours — Example 1

Relevant Experimental Evidence - Example 1: 90/10 HNBR/NBR blend with silica and silane coupling agent was compounded and cured. - Example 2: Increasing NBR fraction to 20 parts raised blend iodine number to 106. - Example 3: 70/30 HNBR/NBR blend gave iodine number 130 and lower tensile strength. - Example 4: 60/40 HNBR/NBR blend gave iodine number 154, showing the upper blend range. - Comparative Example: 50/50 blend served as the high-NBR reference.

Technical Relevance Relevant because it shows how HNBR can be economically diluted with NBR while preserving sealing performance. The patent is useful for understanding how residual unsaturation and blend ratio affect final vulcanizate properties in hydrogenated nitrile rubber systems.

Patent 6

- Patent Number: US7772328B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses low-molecular-weight HNBR prepared by olefin metathesis of nitrile rubber, using ruthenium/osmium catalysts and optionally a co-olefin. The process is aimed at producing HNBR with reduced Mooney viscosity and narrower molecular-weight distribution in a single-step metathesis/hydrogenation concept. - Target molecular weight (MW): 30,000 to 250,000 — Claim 1 - Target Mooney viscosity (ML 1+4 @100°C): less than 10 — Claim 1 - Target polydispersity

index: less than 2.5 — Claim 1 - Target Mw: 40,000 to 220,000 — Claim 2 - Target polydispersity index: less than 2.3 — Claim 3 - Cement concentration: 15 wt.-% solid — Example 1 - Ethylene pressure: 6.9×10^5 Pa — Example 1 - Reactor temperature: 80°C — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Co-olefin: 1-hexene — Example 4 - Co-olefin amount: 150 g — Example 4 - Mooney viscosity (ML 1+4 @100°C): 73 — Starting polymer (Table 2) - Mooney viscosity (ML 1+4 @100°C): 43 — Experiment 1 (Table 2) - Mooney viscosity (ML 1+4 @100°C): 28 — Experiment 2 (Table 2) - Mooney viscosity (ML 1+4 @100°C): 3 — Experiment 3 (Table 2) - Mooney viscosity (ML 1+4 @100°C): 31 — Experiment 4 (Table 2) - Mn: 85 kg/mol — Substrate (Table 2) - Mw: 296 kg/mol — Substrate (Table 2) - PDI: 3.50 — Substrate (Table 2) - Mn: 27 kg/mol — Typical product description - Mw: 54 kg/mol — Typical product description - PDI: 2.0 — Typical product description

Relevant Experimental Evidence - Example 1: Ethylene co-olefin at 80°C with 0.05 phr catalyst reduced Mooney viscosity to 43. - Example 2: Increasing catalyst load to 0.10 phr reduced Mooney viscosity further to 28. - Example 3: 0.25 phr catalyst gave the lowest Mooney viscosity, 3, and PDI 1.90. - Example 4: 1-hexene co-olefin also produced low-Mooney HNBR, confirming co-olefin flexibility.

Technical Relevance Highly relevant because it directly targets low-Mooney HNBR, a key processing challenge in hydrogenated nitrile rubber manufacture. It provides explicit process levers—co-olefin, catalyst loading, and temperature—for controlling molecular weight during HNBR production.

Patent 7

- Patent Number: US9150669B2
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a one-step process for producing low-molecular-weight HNBR by simultaneous hydrogenation and metathesis of nitrile rubber in the presence of hydrogen and a ruthenium/osmium catalyst. The process is intended to deliver HNBR with low residual double bonds, lower molecular weight, and narrower molecular-weight distribution. - Target molecular weight (Mw): 20,000 to 250,000 — Abstract - Target Mooney viscosity (ML 1+4 @100°C): 1 to 50 — Abstract - Target polydispersity index: less than 3.0 — Abstract - Hydrogenation catalyst loading: 0.041 phr — Example 1 - Hydrogenation catalyst loading: 0.045 phr — Example 1 - Hydrogenation catalyst loading: 0.060 phr — Example 1 - Metal content: 65.6 ppm — Example 1 - Co-catalyst: triphenylphosphine — Example 1 - Co-catalyst loading: 1.0 phr — Example 1 - Reaction time: 0 min — Example 1 - Reaction time: 60

min — Example 1 - Reaction time: 120 min — Example 1 - Reaction time: 180 min — Example 1 - Reaction time: 240 min — Example 1 - RDB: 100% at 0 min — Example 1 - RDB: 2.5% at 60 min — Example 1 - RDB: 0.6% at 120 min — Example 1 - RDB: 0.5% at 180 min — Example 1 - RDB: 0.4% at 240 min — Example 1 - RDB: 7.0% at 60 min — Example 1 - RDB: 4.5% at 120 min — Example 1 - RDB: 3.9% at 180 min — Example 1 - RDB: 3.5% at 240 min — Example 1 - RDB: 15.4% at 60 min — Example 1 - RDB: 3.1% at 120 min — Example 1 - RDB: 1.3% at 180 min — Example 1 - RDB: 0.7% at 240 min — Example 1 - Mw: 113371 g/mol — HNBR 1 after 240 min - Mn: 48352 g/mol — HNBR 1 after 240 min - PDI: 2.34 — HNBR 1 after 240 min - Mooney viscosity: 33.3 — HNBR 1 after 240 min - Mw: 133758 g/mol — HNBR 2 after 240 min - Mn: 54595 g/mol — HNBR 2 after 240 min - PDI: 2.45 — HNBR 2 after 240 min - Mooney viscosity: 41.3 — HNBR 2 after 240 min - Mw: 195304 g/mol — HNBR 3 after 240 min - Mn: 64130 g/mol — HNBR 3 after 240 min - PDI: 3.05 — HNBR 3 after 240 min - Mooney viscosity: 69.3 — HNBR 3 after 240 min

Relevant Experimental Evidence - Example 1: Three catalyst systems were compared, showing different hydrogenation rates and final Mooney viscosities. - Example 1: HNBR 1 achieved the fastest and deepest reduction in residual double bonds. - Example 1: HNBR 1 also gave the lowest final Mooney viscosity among the three catalysts. - Example 1: HNBR 3, the industrial norm, retained the highest residual unsaturation and viscosity.

Technical Relevance This is one of the most directly relevant patents because it combines hydrogenation and metathesis to make low-Mooney HNBR in a single step. It provides concrete catalyst and kinetic data showing how to drive both unsaturation reduction and molecular-weight reduction.

Patent 8

- Patent Number: EP2700692B1
- Patent Title: Hydrogenated nitrile rubber composition
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a hydrogenated nitrile rubber composition for static seals, not the synthesis of HNBR itself. The formulation uses HNBR with controlled bound acrylonitrile and low iodine value, combined with carbon black and/or other fillers, a polyfunctional unsaturated cocrosslinking agent, and peroxide to achieve low hardness and excellent LLC resistance. - Hydrogenated nitrile rubber bound AN content: 31 to 50% — Claim 1 - Hydrogenated nitrile rubber iodine value: 30 mg/100 mg or less — Claim 1 - Carbon black and/or other filler: 30 to 300 parts by weight — Claim 1 - Polyfunctional unsaturated cocrosslinking agent: 0.5 to 30 parts by weight — Claim 1 - Organic peroxide: 1 to 10 parts by

weight — Claim 1 - Durometer A hardness: less than 70 — Claim 1 - Hydrogenated nitrile rubber AN content: 36.2% — Example discussion - Degree of hydrogenation: 96% — Example discussion - Iodine value: 11 — Example discussion - MT carbon black loading: 150 parts by weight — Example 1 - Polyfunctional unsaturated compound loading: 6 parts by weight — Example 1 - Organic peroxide loading: 3 parts by weight — Example 1 - Antioxidant loading: 3 parts by weight — Example 1 - Silica loading: 30 parts by weight — Comparative Example 1/7 context - Silica loading: 310 parts by weight — Comparative Example 1/7 context - FEF carbon black particle size: 45 nm — Comparative Example 1 - MT carbon black particle size: 350 nm — Example 1 - Compression molding temperature: 180°C — Example 1 - Compression molding time: 10 minutes — Example 1 - LLC test temperature: 120°C — Example 1 - LLC test duration: 2,000 hours — Example 1

Relevant Experimental Evidence - Example 1: HNBR-A with MT carbon black and cocrosslinking agent was molded into a 2 mm sheet for LLC testing. - Example 2 and 3: Lower and higher MT carbon black loadings were evaluated against the baseline formulation. - Example 7: Silica replaced MT carbon black to assess filler effects on LLC resistance and processability. - Example 15: Combined MT carbon black and silica were used to balance sealing performance and processability. - Example 18 and 19: Silane coupling agent was added with mixed fillers to improve filler interaction.

Technical Relevance Relevant because it defines a seal-oriented HNBR formulation window with low hardness and strong coolant resistance. While not a synthesis route, it provides practical property targets and compounding constraints for hydrogenated nitrile rubber used in mechanical seals.

Patent 9

- Patent Number: US5569719A
- Patent Title: Vulcanizable partially hydrogenated nitrile rubber composition containing ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 1996
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses preparation of partially hydrogenated nitrile-butadiene copolymer rubber by hydrogenating the butadiene units of NBR in methyl isobutyl ketone using a Pd/SiO₂ catalyst. The partially hydrogenated polymer is then sulfur-vulcanized with tellurium dithiocarbamate promoter to reduce heat build-up and compression set. - Bound acrylonitrile content: 36 wt.% — Example 1 - Solvent: methyl isobutyl ketone — Example 1 - Catalyst: Pd/SiO₂ — Example 1 - Iodine value: 15 — Example 1 - Hydrogenation degree: 95% — Example 1 - Vulcanization temperature: 160°C — Example 1 - Vulcanization time: 20 minutes — Example 1 - Sulfur loading: 0.5 parts by weight — Example 1,

recipe a - Sulfur loading: 0.75 parts by weight — Example 1, recipes b/c/d - Sulfur loading: 1.0 parts by weight — Example 1, recipe f - Tetramethylthiuram disulfide loading: 1.5 parts by weight — Example 1 - 2-mercaptobenzothiazole loading: 0.5 parts by weight — Example 1 - Tellurium diethyldithiocarbamate loading: 1.0 parts by weight — Example 1, recipe c/e/f - Tellurium diethyldithiocarbamate loading: 2.0 parts by weight — Example 1, recipe d - Iodine value: 10 — Example 2 - Iodine value: 20 — Example 2 - Iodine value: 30 — Example 2 - Heat build-up: 35°C — Comparative Example 1, Run 1 - Heat build-up: 32°C — Comparative Example 1, Run 2 - Heat build-up: 22°C — Example 1, Run 3 - Heat build-up: 21°C — Example 1, Run 4 - Heat build-up: 24°C — Example 1, Run 5 - Heat build-up: 20°C — Example 1, Run 6 - Heat build-up: 40°C — Example 2, Run 7 - Heat build-up: 27°C — Example 2, Run 8 - Heat build-up: 19°C — Example 2, Run 9 - Heat build-up: 27°C — Example 2, Run 10 - Heat build-up: 19°C — Example 2, Run 11 - Heat build-up: 17°C — Example 2, Run 12

Relevant Experimental Evidence - Example 1: Partially hydrogenated NBR with iodine value 15 was prepared using Pd/SiO₂ and then sulfur-cured. - Example 1: Tellurium dithiocarbamate-containing recipes showed lower heat build-up than recipes without it. - Example 2: Lower iodine values (10 and 20) gave lower heat build-up than iodine value 30. - Example 2: The tellurium promoter further reduced heat build-up across all iodine values.

Technical Relevance This is a foundational HNBR-related patent because it explicitly describes partial hydrogenation of NBR to a low-iodine HNBR precursor. It also links residual unsaturation to cure behavior and thermal performance, which is central to HNBR design.

Patent 10

- Patent Number: EP2519350B1
- Patent Title: Highly active metathesis catalysis selective for romp and rcm
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is primarily about ruthenium metathesis catalysts and their use in metathesis, depolymerization, and hydrogenation of rubbers including NBR and HNBR. For HNBR relevance, it discloses a two-step or sequential process in which NBR is depolymerized by metathesis and then hydrogenated under high pressure at elevated temperature to form depolymerized HNBR. - Ligand synthesis temperature: 85°C — Example 1 - Ligand synthesis yield: 86% — Example 1 - Ru complex preparation temperature: 20-25°C — Example 1 - Ru complex preparation time: 30-60 min — Example 1 - Ru complex yield: 4% — Example 1 - Ligand synthesis scale: 5.0 mmol — Examples 1-19 - Ru complex preparation scale: 1.0 mmol — Examples 1-19 - Depolymerization temperature: 30 to 100°C — Description - Hydrogenation temperature: 60 to 150°C — Description - Hydrogen pressure: high

pressure — Description - Catalyst type: ruthenium or osmium complex — Claim 1 - L1 and L2: chloride anion — Claim 1 - M: ruthenium — Claim 1 - Method includes depolymerization of modified NBR or SBR — Description - Method includes making depolymerized HNBR by depolymerization followed by hydrogenation — Description

Relevant Experimental Evidence - Example 1: A Ru complex ligand and catalyst were synthesized at 85°C and then converted to a green Ru complex. - Example 2: An alternative Ru complex was prepared from a different precursor, showing modular catalyst synthesis. - Example 7: A ligand was confirmed by LC-MS before Ru complex formation. - Example 20: Another Ru complex variant was prepared using the alternative Ru reagent 1a.

Technical Relevance Relevant because it explicitly includes a route to depolymerized HNBR via metathesis followed by hydrogenation, which is a nontraditional but important HNBR processing pathway. It also provides catalyst design and temperature windows for metathesis/hydrogenation chemistry involving nitrile rubbers.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- The patent set splits into synthesis/purification patents (US7919563B2, US7772328B2, US9150669B2, US5569719A, EP2519350B1, US7846995B2) and formulation patents (US9321910B2, US7772305B2, US20090258980A1, EP2700692B1).
- Direct HNBR synthesis patents emphasize solution-phase hydrogenation of NBR, often with homogeneous Rh/Ru catalysts or Pd/SiO₂, while the metathesis-based patents use Ru/Os catalysts and co-olefins to reduce molecular weight before or during hydrogenation.
- Residual unsaturation control is a recurring theme: US5569719A targets iodine value 15; US7772305B2 and US20090258980A1 specify iodine number windows of 32-65; EP2700692B1 requires iodine value 30 mg/100 mg or less; US9150669B2 reports RDB reduction from 100% to below 1% in one catalyst system.
- Catalyst systems differ substantially: Pd/SiO₂ in US5569719A, Ru/Os metathesis catalysts in US7919563B2 and US7772328B2, Ru hydrogenation/metathesis catalysts in US9150669B2, and functionalized ion-exchange resins for Ru removal in US7846995B2.
- Process conditions are most explicit in the metathesis/hydrogenation patents: 80°C and 6.9×10⁵ Pa ethylene in US7919563B2; 80°C and 0.05-0.25 phr catalyst in US7772328B2; 60-150°C under high pressure in EP2519350B1; 66 h at ca. 100°C for Ru removal in US7846995B2.
- Formulation patents consistently use peroxide curing, with optional polyfunctional unsaturated coagents, and filler systems based on carbon black, silica, or mixed fillers to tune hardness, LLC resistance, and processability.
- US9321910B2 is distinct in using microgels plus unsaturated carboxylic acids/salts to improve high-temperature vulcanizate properties, whereas EP2700692B1 focuses on low-hardness static seals with MT carbon black and cocrosslinking agent.

- US20090258980A1 and US7772305B2 both exploit HNBR/NBR blending to reduce cost and adjust iodine number, but US20090258980A1 explicitly blends in unhydrogenated NBR while US7772305B2 centers on HNBR composition windows and plasticizer selection.

4. CONCLUSION

The evidence supports solution-phase hydrogenation of NBR as the core HNBR synthesis route, with selective Rh- or Pd-based hydrogenation giving low residual unsaturation and acceptable molecular weight, and Ru/Os metathesis routes offering an alternative for molecular-weight reduction and processability control. For practical HNBR manufacture, the most defensible parameter windows from the patents are high hydrogenation of diene units (about 95-100% in one formulation patent), low iodine value/residual double bond content, and careful control of catalyst loading and temperature in the 80°C to 150°C range depending on the route. For downstream performance, peroxide curing with optional polyfunctional unsaturated coagents, controlled acrylonitrile content, and filler selection (MT carbon black, silica, or mixed systems) are repeatedly shown to govern oil resistance, heat resistance, low-temperature flexibility, and roll processability. If the goal is a robust HNBR synthesis strategy, the strongest evidence favors homogeneous hydrogenation of NBR followed by catalyst removal, with metathesis-assisted molecular-weight reduction as a useful adjunct when low Mooney viscosity is required.

5. REFERENCES

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- US7919563B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7919563B2/en>
- US7772305B2 | Hydrogenated NBR composition | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7772305B2/en>
- US7846995B2 | ... -containing catalyst residues from optionally hydrogenated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7846995B2/en>
- US20090258980A1 | Hydrogenated nbr composition | Not disclosed | US | 2009 | <https://patents.google.com/patent/US20090258980A1/en>
- US7772328B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7772328B2/en>
- US9150669B2 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9150669B2/en>
- EP2700692B1 | Hydrogenated nitrile rubber composition | Not disclosed | EP | 2019 | <https://patents.google.com/patent/EP2700692B1/en>
- US5569719A | Vulcanizable partially hydrogenated nitrile rubber composition containing ... | Not disclosed | US | 1996 | <https://patents.google.com/patent/US5569719A/en>

- EP2519350B1 | Highly active metathesis catalysis selective for romp and rcm | Not disclosed | EP | 2019 | <https://patents.google.com/patent/EP2519350B1/en>